

Spectral concentration in Vina docking-score matrices: what two-way centering reveals and does not correct

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Abstract

Reverse docking, off-target profiling and multiobjective design use docking scores as a ligand-by-target matrix, yet the statistical structure of this matrix can confound shared score tendencies with target-resolved information. We measured spectral concentration in two large AutoDock Vina matrices using the participation-ratio (PR) effective dimension of the target-correlation spectrum. A 12,651-compound by 44-target panel had PR dimension 1.83, and DOCKSTRING, with 260,060 complete molecules by 58 targets, had PR dimension 2.27. Removing ligand and target main effects changed the estimand and increased these values to 9.30 and 18.21. Chemical-cluster resampling and independently seeded chemical supports confirmed the increase, while fitted additive-noise nulls showed that the residual surfaces remained concentrated. On a dense matched 74-compound by six-target ChEMBL surface, the experimental-minus-docking PR difference was 2.01 (paired ligand-bootstrap interval 1.17–2.67) before centering but 0.48 (−0.07–0.92) afterward. We then expanded the operational comparison to 137 ligands, 38 targets and 2,522 non-tied observed experimental target pairs without activity imputation. Absolute, column-standardized and two-way-residual Vina reached mean per-ligand pairwise accuracies of 0.566, 0.546 and 0.564, with no resolved paired difference. An external experimental target prior reached 0.525; a non-deployable prior fitted with each ligand’s scaffold cluster held out reached 0.592, revealing observation-cohort structure. Rankings changed when more extensively profiled ligands were required. Two-way centering is therefore useful for diagnosing residual score structure, but higher residual dimension is not by itself a validated correction for target ranking.

Scientific Contribution. We provide a preprocessing-matched spectral analysis of two large dense Vina panels and separate uncertainty in chemical and target-panel composition from

transformation effects. A dense matched spectral control and an expanded observed-pair experimental benchmark show both what two-way centering reveals—higher-dimensional residual structure—and what it does not establish: improved target ranking.

Keywords: molecular docking; AutoDock Vina; effective dimension; spectral analysis; two-way centering; reverse docking; target profiling.

1 Introduction

A docking score is scalar for one protein–ligand pose, but many applications produce a matrix. Reverse docking ranks targets for a ligand, secondary-pharmacology panels compare off-target liabilities, and multiobjective design uses scores against several desired and undesired proteins. In each case the useful object is a matrix $X \in \mathbb{R}^{N \times P}$ whose rows are ligands and columns are targets. Interpreting its P columns as P independent target-resolved descriptors assumes that the matrix actually contains a commensurate number of independent directions.

Two long-recognized biases make this assumption doubtful. Empirical energy functions often reward larger ligands because additional atoms contribute additional favorable contacts; normalization by molecular weight, heavy-atom count or learned physicochemical corrections can change virtual-screening rankings.[4, 5, 6] Across proteins, scores also depend on pocket contact area, size and hydrophobicity. In a reverse-docking study of 5,000 molecules and 100 proteins, these target properties produced recurrent “interference proteins” and motivated per-target normalization.[7] These observations identify ligand and target main effects. They do not say how much of a complete score matrix those effects occupy, how many collective directions remain, or whether a low-dimensional column-standardized surface is equivalent to poor selectivity.

Two-directional normalization and principal-component analysis of docking matrices also precede this study. SePreSA applied a ligand-wise Z transformation followed by normalization of protein-score vectors to a 79-compound by 86-protein docking matrix. [1] Fukunishi and colleagues applied PCA to a complete docking matrix for compound and pocket classification, and later used multi-protein docking-score profiles as QSAR descriptors.[2, 3] These studies establish that matrix-wide normalization and PCA are useful. The contribution here is narrower: a large-scale quantitative estimate of spectral concentration, explicit comparison of column-standardized and two-way-centered estimands, and uncertainty analyses for chemical and target-panel composition.

The distinction matters because docking has several separable tasks. A method may reproduce a crystallographic pose yet rank affinities poorly, and no single program performs uniformly across targets and evaluation objectives.[8] Likewise, the aggregate variance of a score matrix and the correctness of within-ligand target contrasts are different objects. Writing

$$X_{ij} = \mu + a_i + b_j + \gamma_{ij}, \quad (1)$$

separates the grand mean μ , the ligand tendency a_i to score well across targets, the target tendency b_j to score ligands favorably, and the interaction γ_{ij} . The column-standardized spectrum is allowed to be dominated by a_i . Selectivity, however, depends on differences within rows and ultimately on γ_{ij} ; the additive terms cancel from a double difference. A PR estimate on X and the same estimate on γ therefore answer different questions. Established panel selectivity summaries such as Gini, entropy and window scores also answer different questions and can reorder the same compounds, making the metric definition part of the estimand rather than a cosmetic choice.[23, 24]

Complete public score matrices now permit this distinction to be tested at useful scale. DOCKSTRING contains docking scores for more than 260,000 molecules against 58 medically relevant targets.[9] An independent acute-toxicity resource contains 12,651 analyzable compounds docked against a 44-protein secondary-pharmacology panel.[10] We combine these matrices with matched experimental affinity data. Two small RF-Score rescoring blocks and learned-affinity examples are retained as illustrative sensitivities in the Supplement because neither provides a large independent replication of the principal Vina result.[11]

Low-rank representations are not new to chemogenomics. Kernelized Bayesian matrix factorization, probabilistic matrix factorization and neighborhood-regularized latent-factor models deliberately embed sparse DTI observations in lower-dimensional spaces to improve prediction or matrix completion.[17, 18, 19] Our question is different: we do not impose a low-rank predictive model on sparse experimental data. We measure the empirical spectrum of dense docking output, then ask which part is attributable to additive ligand and target effects. The two-way decomposition connects this diagnostic to classical additive-model reasoning, while participation and entropy ranks provide continuous summaries rather than a chosen latent dimension.[20, 21]

We ask four linked questions. Do the two large tested Vina panels concentrate their variance into a few PR effective dimensions? Is that observation stable to chemical and target-panel

composition? What remains after ligand and target main effects are removed? Finally, does a higher-dimensional residual representation improve the operational task of ranking targets for a ligand? The last comparison is necessary because a transformation can clarify an estimand without becoming a better scoring rule. Our evidence supports two-way centering as a spectral diagnostic, not as an automatically beneficial correction for reverse docking.

2 Results

2.1 Two independent Vina panels have concentrated column-standardized spectra

The 12,651-by-44 antitarget panel was analyzed after clipping the rare positive Vina scores to zero, mean-imputing missing entries and standardizing each target column. The largest eigenvalue accounted for 73.3% of the correlation spectrum. The PR effective dimension,

$$r_{\text{PR}} = \frac{(\sum_k \lambda_k)^2}{\sum_k \lambda_k^2}, \quad (2)$$

has a direct correlation interpretation. Because a P -target correlation matrix has $\text{tr}(C) = P$,

$$r_{\text{PR}} = \frac{P^2}{\text{tr}(C^2)} = \frac{P}{1 + (P-1)\overline{r_{ij}^2}}, \quad (3)$$

where the bar is over unique off-diagonal target pairs. Thus, a small PR dimension is an alternative summary of strong mean-squared inter-target correlation; the full spectrum additionally shows how that dependence is distributed among collective modes. The estimated PR dimension was 1.834, with a Butina-cluster bootstrap interval of 1.783–1.884 (Fig. 1). Three leading modes exceeded both the descriptive rank-wise 95% column-permutation threshold and a studentized maximum-deviation envelope controlling the family-wise error rate over eigenvalue ranks. The matrix is not algebraically rank two: eight principal components are needed to reach 90% of variance, and the entropy effective rank is 3.90. “Collapse toward two” therefore refers specifically to participation-ratio variance concentration. Full column-standardized and residual scree plots are given in Supplementary Fig. S1.

The 111 zero-censored cells did not drive this concentration. Treating them as missing and target-mean imputing gave a PR dimension of 1.829, while excluding every affected ligand gave 1.814 (12,544 ligands). Excluding the four most-censored targets gave 1.602, but this changes the target estimand and is interpreted as panel-composition sensitivity rather than an

imputation effect (Supplementary Table S2). Complete-case analysis gave 1.320 on 9,297 ligands; because this selects a different chemical population, it is a support sensitivity, not a like-for-like missing-data correction. A chemical-cluster block-permutation null and a Marchenko–Pastur edge based on the 8,150 effective clusters both retained three leading modes, matching the independent-column null.

The independent DOCKSTRING matrix gave the same qualitative result at much larger N and with 58 targets. Among 260,060 molecules with complete scores, principal component 1 accounted for 65.9% of the spectrum and r_{PR} was 2.266 (molecule-bootstrap interval 2.239–2.298). Five independently seeded 15,000-molecule supports gave point estimates from 2.252 to 2.294; support-specific Bemis–Murcko scaffold-cluster intervals are shown in Supplementary Fig. S6. Four leading modes exceeded both the rank-wise and simultaneous common-protocol null thresholds. Its entropy rank was 6.21 and 22 components were needed for 90% of variance, again distinguishing a concentrated spectrum from literal rank deficiency. Retaining the 5,393 positive scores rather than clipping them changed the raw PR dimension only from 2.266 to 2.275. These intervals and support comparisons remain conditional on the fixed 58-target panel.

The result was also stable to the observed target-panel composition. Leave-one-target-out PR dimensions ranged from 1.756 to 1.853 for Docking-44 and from 2.218 to 2.311 for DOCKSTRING-58 (Supplementary Fig. S2). Random target subsets showed saturation rather than a linear increase: at 20 targets the median column-standardized PR dimensions were 1.79 and 2.18, compared with 1.83 and 2.27 for the full panels (Fig. 1c). Broad target-family-stratified sampling gave similar 20-target medians of 1.85 and 2.22. For the two-way-centered residuals, the corresponding family-stratified medians were 7.38 and 11.40, increasing to 9.30 and 18.21 in the full panels (Fig. 1d). These subset distributions quantify sensitivity within the tested biological panels; the target lists are not probability samples from a universal target population.

2.2 The leading direction is a shared docking-score axis

The dominant direction was shared across targets rather than localized to one receptor family. After orienting principal component 1 (PC1) consistently, 43 of 44 Docking-44 loadings and 57 of 58 DOCKSTRING loadings were positive (Fig. 2c,d). The PC1 scores correlated 0.997 and 0.995 with the corresponding per-ligand mean of the original score matrix. These are raw score-scale row means, not means calculated after column standardization. Seven physicochemical descriptors explained 60.6% of the variance in the Docking-44 row mean; heavy-atom count

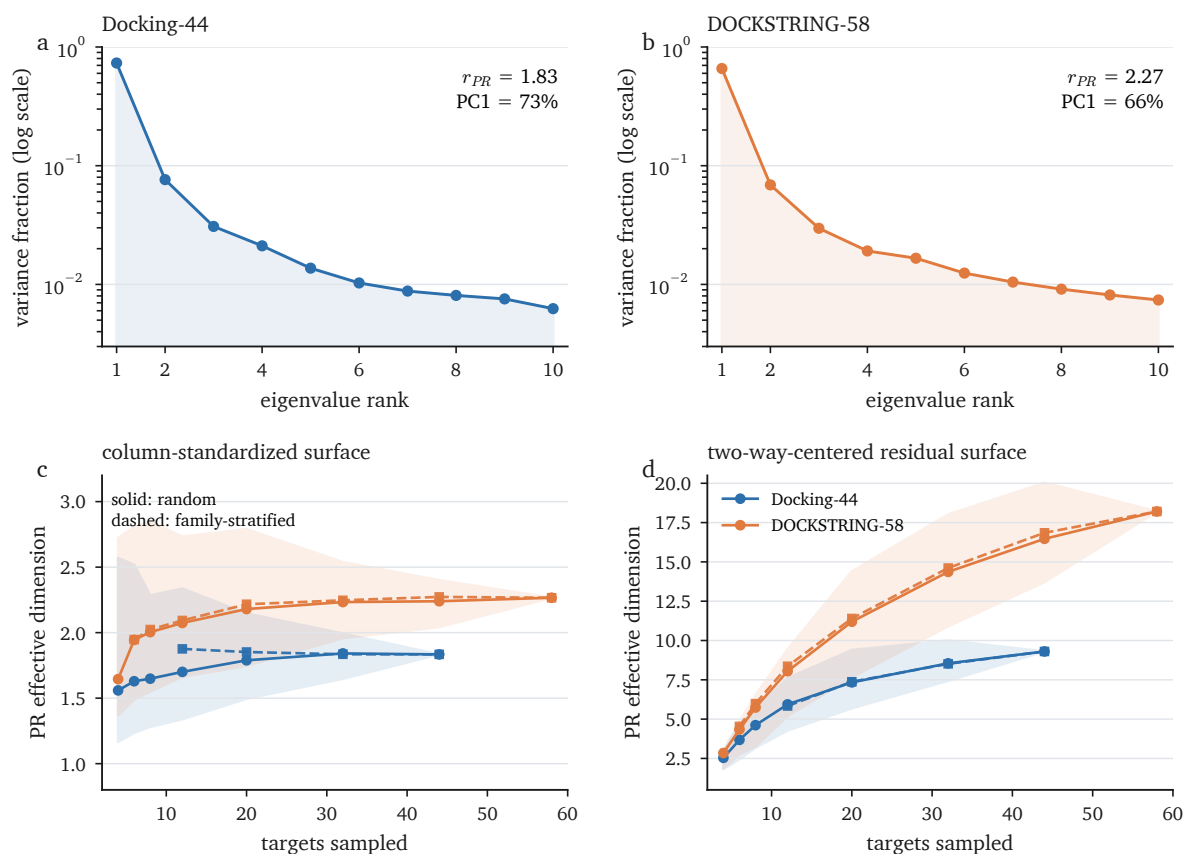


Figure 1: Two independent Vina panels have concentrated column-standardized spectra. a,b, Eigenvalue spectra and PR effective dimensions of the 44-target antitarget and 58-target DOCKSTRING matrices after column standardization. **c,d,** Column-standardized and two-way-centered residual PR effective dimensions under target subsampling. Solid lines and shaded 2.5–97.5% intervals show 200 uniform subsets; dashed square-marked lines show 200 family-stratified subsets where the requested size permits representation of every broad family. Full-panel points use all ligands.

alone explained 18%.

Removing size-related information weakened but did not erase the concentration. Linear residualization of each target score against seven physicochemical descriptors increased r_{PR} from 1.83 to 2.54; cross-fitted nonlinear residualization increased it to 3.52. Restricting the data to 4,308 molecules with 17–22 heavy atoms still gave PR dimension 1.70. Dividing scores by heavy-atom count made the spectrum more concentrated (PR dimension 1.14), showing that ligand-efficiency normalization is not a dimensionality correction. Across 36 targets, pocket volume correlated 0.514 (target bootstrap interval 0.272–0.686) with the absolute correlation between each target and the per-ligand panel mean. A 13-descriptor pocket model had negative leave-one-target-out R^2 for both ordinary least squares (-0.011) and ridge regression (-0.144), so the data do not support a multivariable pocket-mechanism claim. On the exact 74-by-6 experimental support, linear and five-fold cross-fitted nonlinear descriptor residualization raised docking PR dimension only to 2.81 and 2.68, respectively, still below the exact-relation median experimental value of 3.83. Thus the matched contrast is not solely a consequence of the seven measured bulk descriptors.

2.3 The docking–experiment contrast narrows after centering

A concentrated docking surface could reflect its compounds or targets rather than its score model. We therefore used the same 74 compounds and six aminergic targets for docking and ChEMBL. The primary experimental curation required an exact standard relation, retained K_i , K_d , IC_{50} and EC_{50} , aggregated repeated compound–target activities by the median, and required measurements for at least five of six targets. Target-median imputation filled 9.9% of cells. The resulting ChEMBL surface had PR dimension 3.833, compared with 1.827 for docking on the same support (Fig. 3a). The paired ligand-bootstrap interval for the experimental-minus-docking difference of 2.006 was 1.174–2.673.

The comparison changed in the residual space. After two-way centering, the docking and experimental PR dimensions were 3.646 and 4.128. Their difference was 0.483, with a paired bootstrap interval of -0.071 – 0.921 (Fig. 3b). Thus, much of the raw docking–experiment contrast on this support arose from the shared ligand direction. It did not persist as a clearly resolved difference between residual effective dimensions.

Multiple imputation supported the same qualitative reading. Across 100 posterior chained-regression imputations, the median experimental PR dimensions were 3.823 before centering and

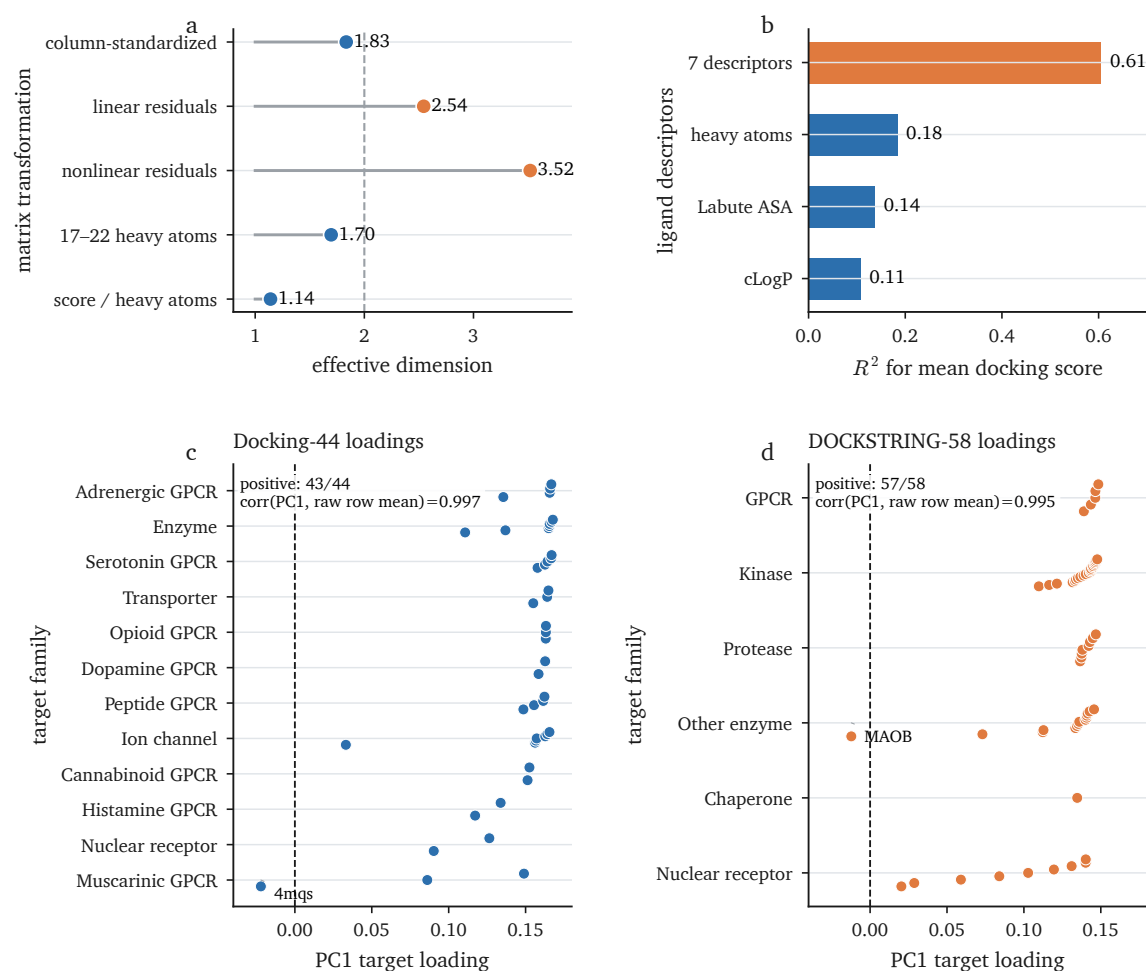


Figure 2: The leading component is a shared docking-score axis. **a**, Docking-44 PR effective dimension after physicochemical transformations; the horizontal segments connect each estimate to one and are visual guides, not confidence intervals. **b**, Variance in the raw per-ligand mean score explained by seven ligand descriptors and single-descriptor controls. **c,d**, PC1 target loadings grouped by curated target family. Loading signs were oriented so that the majority are positive. The annotations report the number of positive loadings and the correlation between PC1 scores and raw per-ligand mean scores. Descriptor residualization is diagnostic and is not proposed as a scoring function.

173 4.069 afterward. The corresponding experimental-minus-docking differences had medians 1.996
174 and 0.423. The displayed 95% ranges across imputations quantify missing-value sensitivity; they
175 are not bootstrap confidence intervals.

176 More restrictive assay blocks preserved the ordering. Human binding assays with all four
177 endpoints gave 3.510 for ChEMBL and 1.865 for docking on 47 ligands. Restricting further to
178 human binding Ki/Kd gave 3.072 and 1.894 on 35 ligands. These comparisons do not define
179 the dimension of experimental binding in general: the assays come from multiple laboratories
180 and conditions, up to 13.3% of cells are imputed, and six targets cap the centered algebraic
181 dimension at five. They support only the statement that these matched ChEMBL surfaces are less
182 spectrally concentrated than docking on the same ligand–target support.

183 Additional controls addressed assay choice, missingness and measurement heterogeneity. A
184 fully observed 31-by-6 ChEMBL sub-block had PR dimension 3.61. On a secondary 212-by-22
185 block, estimates were 5.72 with K-nearest-neighbour imputation, 8.59 averaged over chained
186 imputations, 2.64 with low-rank matrix completion and 7.28 with probabilistic-PCA expectation
187 maximization; the magnitude depended on the estimator. A separate frozen ChEMBL extraction
188 with the same nominal 212-by-22 shape and 52.9% missingness gave KNN PR dimension 6.85; we
189 retain both in Supplementary Table S3 as a curation-sensitivity discrepancy. A legacy most-potent
190 aggregation of the 74-by-6 block gave 3.734 for experiment and 1.827 for docking, with a paired
191 bootstrap difference interval of 1.05–2.61, and is retained only as a sensitivity. Different-document
192 repeats estimated a standard deviation of 0.493 pChEMBL units. Adding target-scaled Gaussian
193 noise at that level to docking raised its median PR dimension to 2.29 (simulation interval 2.12–
194 2.50); approximately 2.5 times that scale was required to reach the legacy experimental value.
195 None of these controls removes all biological, laboratory or condition-related heterogeneity.

196 **2.4 Centering reveals higher-dimensional but still concentrated residual surfaces**

197 The column-standardized spectrum mixes the ligand main effect with the ligand–target residual.
198 We removed both row and column means according to Eq. 1 and then applied the same col-
199 umn standardization and spectral estimator. This changed the measured object and the answer
200 (Fig. 4a). Docking-44 rose from PR dimension 1.834 to 9.303; a paired Butina-cluster bootstrap
201 placed the increase at 6.982–7.909. DOCKSTRING rose from 2.266 to 18.206, and five indepen-
202 dent 15,000-molecule supports gave increases from 15.880 to 16.045. The corresponding PC1
203 fractions fell from 73.3% to 27.4% and from 65.9% to 17.8%. In unstandardized sum-of-squares

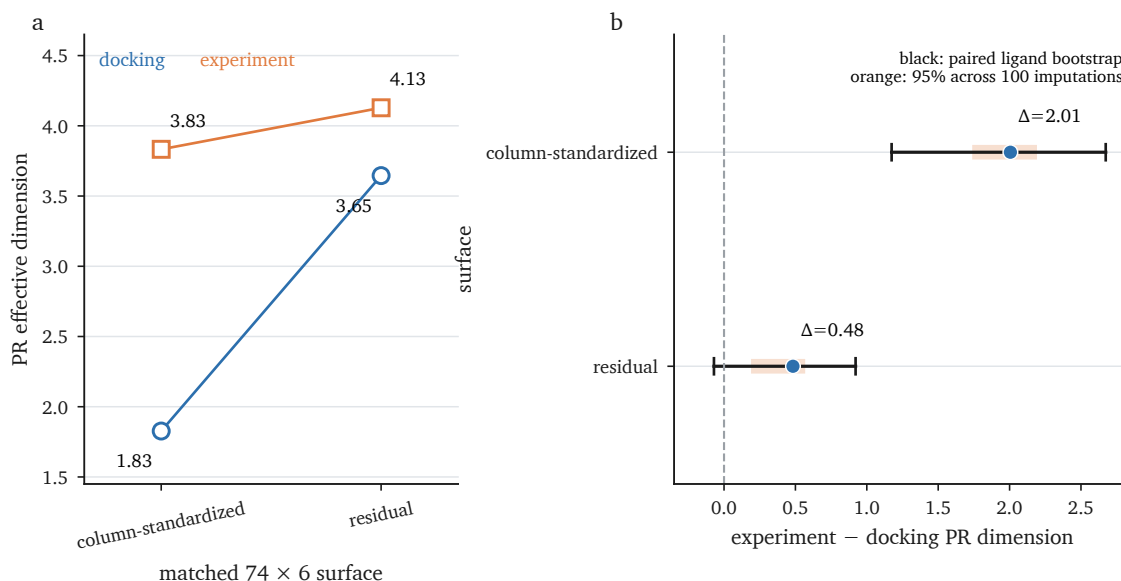


Figure 3: The matched docking-experiment contrast is mainly a raw-surface contrast. **a**, PR effective dimensions for docking and exact-relation median ChEMBL on the same 74 compounds and six targets, before and after two-way centering. **b**, Experimental-minus-docking differences. Black intervals are paired ligand-bootstrap 95% intervals; orange bands are the central 95% of the difference across 100 posterior multiple imputations. These intervals answer different uncertainty questions and are not pooled.

terms, the residual retained 27% and 31% of grand-centered variation. Low variance relative to a main effect is not absence.

The increase was not only a rescaling artifact of the PR denominator. Relative to the available algebraic maximum, normalized PR dimensions increased from 0.042 (1.834/44) to 0.216 (9.303/43) for Docking-44 and from 0.039 (2.266/58) to 0.319 (18.206/57) for DOCKSTRING. This normalization is a ceiling reference, not a correction for target count: the residual PR dimension increased with panel size while its ratio to $P - 1$ decreased (Fig. 1d). At a matched 44 targets, the DOCKSTRING median residual PR dimension was 16.47 (target-subset interval 13.72–20.03), compared with 9.30 for the full Docking-44 panel. The cross-dataset difference therefore remained at matched P , although part of the 18.21-versus-9.30 contrast reflects different target counts.

Higher-dimensional did not mean unconcentrated. We fitted an additive null retaining the observed grand mean, ligand effects, target effects and target-specific residual variances, but drawing residuals independently. After two-way centering, its median PR dimensions were 42.40 for Docking-44 and 56.48 for DOCKSTRING, close to the respective $P - 1$ ceilings of 43 and 57. The observed residual values, 9.29 and 18.16 on the same 12,000-row supports, were smaller than every one of 500 null realizations (one-sided empirical $p = 0.002$ for each; Fig. 4b).

Centering therefore exposes more residual directions, while the residual correlations remain far stronger than independent-noise expectations.

In the transformation-matched permutation analysis, the descriptive rank-wise mode counts increased from 3 to 7 and from 4 to 10. The same counts survived the simultaneous studentized maximum-deviation envelope and were unchanged in five independent 100- permutation series. Because even weak correlations can become statistically detectable at large sample size, we use these counts to describe spectrum shape rather than as a measure of practical dimension.

Two-way centering also creates an exact structural constraint: every interaction row sums to zero. Non-zero column scaling preserves one linear dependence, so the interaction matrix has algebraic rank at most $P - 1$ and one correlation eigenvalue is zero. Residual and column-standardized PR dimensions therefore describe different constrained spaces; the centered increases occur despite, not because of, a larger algebraic maximum.

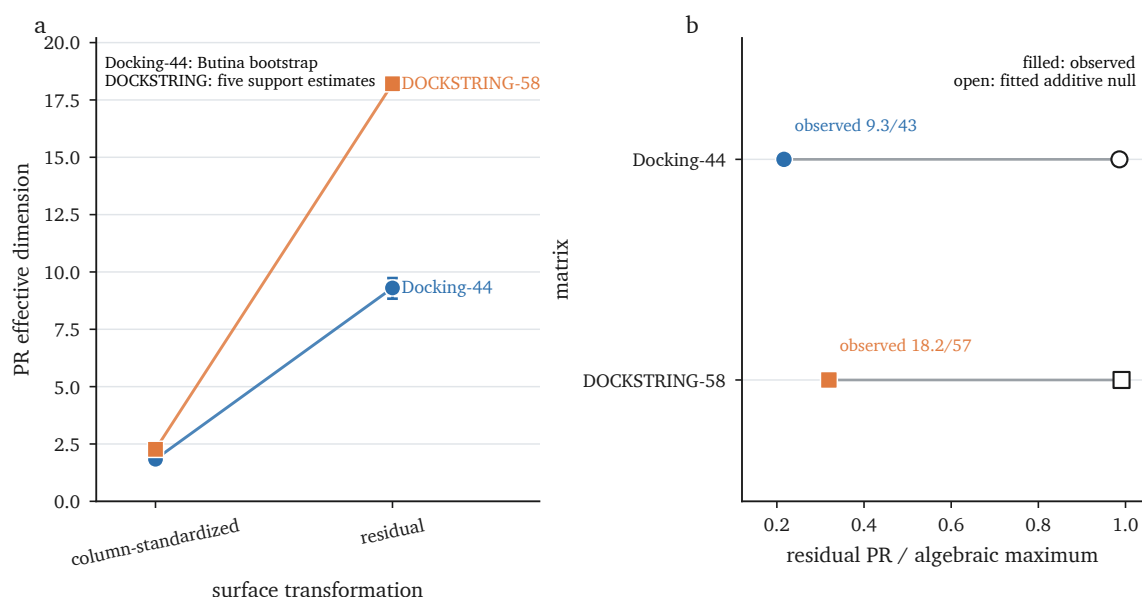


Figure 4: Two-way centering increases effective dimension without removing residual concentration. a, Paired raw-to-residual changes. Docking-44 error bars are 95% Butina-cluster bootstrap intervals; DOCKSTRING error bars span point estimates across five independently seeded 15,000-molecule supports. b, Observed residual PR dimensions and fitted additive-null medians, each divided by its $P - 1$ algebraic maximum. Open-marker bars are the central 95% of 500 null realizations and are visually narrower than the marker. The null preserves fitted ligand and target effects and target-specific residual variance, but removes cross-target residual dependence.

2.5 Higher residual dimension does not by itself improve target ranking

We expanded the operational endpoint beyond the dense six-target block. Exact full-InChIKey matching identified 276 Docking-44 ligands with at least one exact-relation K_i , K_d , IC_{50} or

EC50 measurement in ChEMBL. Requiring at least two observed targets retained 137 ligands, 38 targets, 691 experimental cells and 2,522 non-tied within-ligand target pairs. No experimental activity was imputed. The primary mean gave every ligand equal weight; predicted score ties received half credit, and its 95% interval resampled 95 Bemis–Murcko scaffold clusters. All docking offsets, scales and imputation means were fitted after excluding the 137 evaluation ligands from the 12,651-ligand reference panel; 89 of their 6,028 docking scores were imputed from these external target means.

Absolute Vina reached mean per-ligand pairwise target-preference accuracy 0.566 (scaffold-bootstrap interval 0.502–0.623; Fig. 5a). Two-way-residual and column-standardized scores reached 0.564 (0.500–0.626) and 0.546 (0.483–0.603). The paired residual-minus-column difference was 0.017 (–0.022–0.054), and the residual-minus-absolute difference was –0.002 (–0.065–0.062). Thus, the point estimates do not resolve a practical advantage for any score transformation. Ligand-identity permutation placed the absolute and residual results above their nulls ($p = 0.014$ and 0.024), and within-ligand permutation of the experimental order gave $p = 0.009$ and 0.012 ; these tests assess association, not superiority over another representation.

Target priors exposed a more consequential boundary. A target order fitted on the 12,514 non-evaluation docking ligands reached 0.525, and an external ChEMBL target prior fitted after excluding the evaluation ligands reached 0.525. Their point estimates were below residual Vina, although the paired residual-minus-prior intervals included zero. By contrast, a target prior within the broadly profiled evaluation cohort, fitted with the held-out ligand’s entire scaffold cluster excluded, reached 0.592 (0.531–0.656). It exceeded residual Vina by 0.028 on average, but the paired interval (–0.127–0.066) was wide. This cohort prior is not a deployable score; it diagnoses target and observation-process structure among these multiply assayed ligands.

Coverage sensitivity changed the score-based ordering (Fig. 5b). Requiring at least three observed targets retained 93 ligands: absolute Vina reached 0.587 (0.537–0.638), residual Vina 0.530 (0.480–0.574), the docking target prior 0.590 (0.542–0.637) and the cohort prior 0.636 (0.576–0.693). At thresholds of five and six observed targets, absolute accuracy remained 0.581 and 0.587 while residual accuracy was 0.540 and 0.537. These nested, smaller strata have broad pointwise intervals, but they show that the inclusive ordering depends in part on the 44 ligands observed for exactly two targets. Restriction to human binding assays gave residual and cohort-prior accuracies 0.581 and 0.581 on 128 ligands; the paired interval was –0.089–0.092. On the 107-ligand human binding Ki/Kd block, they were 0.579 and 0.608, with an interval of

−0.131−0.075. Neither residual-minus-column interval excluded zero (Supplementary Table S5). Removing cells whose repeated measurements spanned more than one pChEMBL unit gave residual accuracy 0.560 on 1,912 pairs.

This comparison supplies the operational boundary of the spectral analysis. Two-way centering answers whether ligand and target main effects conceal structured residual variation. It does not determine whether that variation orders targets correctly, and the sparse ChEMBL observation process can make target priors stronger than molecule-specific docking contrasts. A reverse-docking workflow should therefore report target-prior and ligand/outcome-permutation baselines, stratify performance by experimental coverage and choose its score transformation by external task-level validation, not by effective dimension alone.

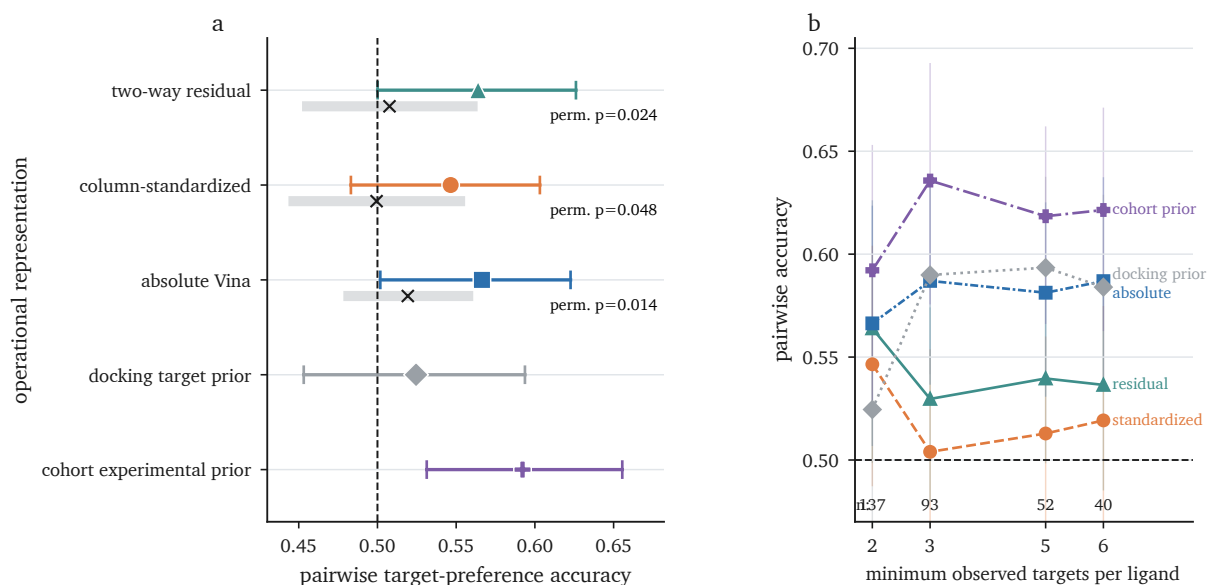


Figure 5: Spectral and operational conclusions are distinct. **a**, Mean per-ligand pairwise target-preference accuracy over 2,522 observed, non-tied experimental target pairs from 137 ligands and 38 targets. Colored intervals are 95% Bemis–Murcko scaffold-cluster bootstrap intervals. Grey bars and crosses show the central 95% and mean of 5,000 ligand-identity permutations for score-based representations; labels give one-sided empirical probabilities. The cohort experimental prior is fitted with the evaluation ligand’s entire scaffold cluster held out and is a diagnostic baseline rather than a docking score. **b**, Accuracy after requiring each ligand to have measurements for at least two, three, five or six targets; vertical lines are pointwise 95% scaffold-bootstrap intervals and labels give the number of retained ligands. The strata are nested rather than independent. The dashed line is chance.

3 Discussion

The two large Vina panels share a concentrated column-standardized spectrum, but centering does not make the residuals unconcentrated. The raw PR dimensions of 1.83 and 2.27 rose to 9.30

and 18.21 after ligand and target main effects were removed; the fitted additive nulls, however, placed independent residual noise near 42 and 56 dimensions. The empirical residuals therefore contain more directions than the shared-score surface while remaining strongly correlated. The result is a reduction in spectral concentration, not its disappearance.

Two-way normalization and PCA themselves predate this work. SePreSA used two-directional normalization of docking-score matrices, and earlier protein–compound affinity work used principal components as multidimensional descriptors.[1, 2, 3] Our contribution is the quantitative separation of raw and residual spectral estimands on two large dense panels, uncertainty with respect to target and chemical composition, and an operational test of whether the richer residual spectrum improves target ranking.

The loading analysis explains why the estimands diverge. Almost every target loaded in the same direction on PC1, and PC1 scores were nearly identical to the raw per-ligand mean score. Ligand bulk accounted for part, but not all, of this direction. Column standardization removes target offsets while retaining the shared ligand tendency; two-way centering removes that tendency and reveals the remaining cross-target covariance. We call this a shared docking-score axis rather than a general-binding axis because neither its biological meaning nor universal validity was established.

On the matched ChEMBL support, experimental affinities were less concentrated than docking before centering, but the paired residual-space difference was smaller and its bootstrap interval included zero. This does not show that Vina residuals reproduce experimental geometry. It shows that the raw docking–experiment contrast on this support is largely a contrast in shared main effects rather than a resolved difference in residual PR dimension.

Operational validation changed the interpretation again. On the expanded 137-ligand, 38-target observed-pair benchmark, absolute and two-way-residual Vina had nearly identical score-based point estimates, and no paired score-transformation contrast was resolved. Both exceeded identity- and outcome-permutation nulls on the inclusive support, but the ordering reversed toward absolute Vina when ligands with only two observed targets were excluded. The deployable external docking and ChEMBL target priors performed below the score-based point estimates. A scaffold-held-out prior from the multiply assayed cohort was comparable or stronger, although it is not a deployable replacement for docking and became indistinguishable from residual Vina in the human binding sensitivity. Precisely because it reflects assay selection, this prior demonstrates how a ranking result can depend on which targets were measured in a

broadly profiled cohort. Higher residual dimension is therefore neither necessary nor sufficient for a useful reverse-docking score.

Accordingly, a spectral audit should report both the column-standardized and two-way-centered spectra, transformation-matched nulls and the chemical and target populations represented by its uncertainty. A reverse-docking benchmark should additionally include target priors, ligand-identity and within-ligand outcome-permutation nulls, and performance stratified by experimental coverage. The operational score representation should be selected on held-out experimental target preferences. Two-way centering can be retained as a diagnostic even when it is not the selected ranking rule.

Both large matrices use Vina-family scores, and the small RF-Score blocks in Supplementary Fig. S3 cannot represent modern learned scoring functions. Neither target panel is a probability sample from all pharmacologically relevant proteins, so jackknife and family-stratified subsampling quantify stability within the observed panels. The dense docking-experiment spectral control still contains only 74 ligands and six aminergic targets, with 9.9% imputed experimental cells. The expanded operational benchmark covers 38 targets but is sparse and non-random: pairwise target co-observation has a median support of only three ligands, and its primary result includes 44 ligands observed against exactly two targets. Scaffold resampling, assay restriction, measurement-dispersion filtering and coverage thresholds expose but cannot remove ChEMBL’s assay-selection and laboratory heterogeneity. The benchmark therefore supports an observed-target ordinal claim, not unmeasured-target retrieval. Finally, the public package reproduces analysis from frozen matrices; it does not rerun upstream receptor preparation, docking, pose generation or RF-Score rescoring.

4 Conclusions

Two-way centering reveals substantially higher-dimensional residual structure in two spectrally concentrated Vina matrices, but the residuals remain far more concentrated than an additive independent-noise null. On dense matched ChEMBL data, the transformation narrowed the docking-experiment PR contrast. On a broader observed-pair benchmark, it did not show a resolved target-ranking advantage over either column standardization or absolute Vina, and the score-based ordering changed with experimental coverage. The practical consequence is to separate diagnosis from correction: spectral analysis can identify shared ligand and target

effects, whereas a score transformation for reverse docking must earn its use against target-prior, permutation and coverage-stratified baselines on held-out experimental data.

5 Methods

5.1 Datasets and score panels

The antitarget matrix was taken from the open resource of Nikitin et al.[10] and contained 12,651 analyzable compounds and 44 Vina-GPU target scores.[16] The source pipeline clipped positive scores to zero; its processed matrix contained 111 such zero-censored cells (0.020%) and no remaining positive values. Missing entries comprised 1.466% of cells and were replaced by the target mean for the primary analysis. Chemical-cluster uncertainty used the precomputed 8,150 Butina clusters supplied with the dataset. A three-target Docking-47 expansion is reported only as an illustrative sensitivity in the Supplement.

DOCKSTRING was read from its released full score table.[9] We used the 58 target columns, clipped positive scores to zero for consistency and used the 260,060 rows with complete scores for the primary spectrum. Only 0.0022% of entries were missing and 0.0358% were positive before clipping; an unclipped sensitivity retained these values.

RF-Score v1 was evaluated with the Open Drug Discovery Toolkit implementation.[11, 14] The first panel comprised the 41 ligands complete across RF-Score, smina/Vina and original Vina scores for 14 targets. The second comprised 35 ligands complete across RF-Score on Vinardo poses and original Vina scores for 20 targets. These are scoring-function and pose-source sensitivities, not affinity benchmarks, and are reported with the Docking-47 expansion in Supplementary Fig. S3.

Experimental activities came from ChEMBL release 34.[22] We retained records with a pChEMBL value and standard type Ki, Kd, IC50 or EC50; canonical SMILES were converted to full InChIKeys and targets were mapped to the panel's single-protein accessions. The primary matched block was rebuilt from activity-level records, required an exact standard relation and full InChIKey, and aggregated repeated compound-target activities by the median. It used 74 compounds with docking scores for six targets and ChEMBL measurements for at least five targets; 9.9% of experimental cells were median-imputed within target. We additionally restricted to human binding assays, either with all four endpoints or with Ki/Kd only. A secondary 212-by-22 block required at least eight measured targets per ligand and 30 measurements per retained

target. We compared five-nearest-neighbor imputation, multiple imputation by chained equations, low-rank matrix completion and probabilistic-PCA expectation maximization. Assay sensitivity separately retained binding records with Ki or Kd endpoints from the already human-only curated refetch. Relations reported as inequalities and records without pChEMBL were excluded. A frozen table that retained the most potent pChEMBL value per compound–target pair was analyzed only as a legacy sensitivity with a paired ligand bootstrap.

For the expanded operational benchmark, the same exact-relation activity-level curation was matched by full InChIKey to all 12,651 Docking-44 ligands before imposing a dense-block criterion. Among 276 ligands with any matched activity, 137 had observations for at least two targets. These ligands contributed 691 observed cells across 38 targets; missing cells were left missing because the endpoint used only experimentally observed within-ligand target pairs. Restriction to human binding assays and then to human binding Ki/Kd defined assay sensitivities. The original 74-by-6 block was retained for the dense same-support spectral comparison and its multiple-imputation sensitivity, not as the primary operational support.

DAVIS experimental affinity[13] and Boltz-2[12] were read from versioned matrices in the release only for a contextual centering ladder in Supplementary Fig. S4. The common surface contained 72 ligands and 442 kinases. PR dimensions were computed on each complete dense matrix, including pKd-floor values, because the surfaces were aligned. The exploratory rank–selectivity analysis is reported entirely in the Supplement; its complete panel has 20 arms, while the 12-arm above-chance subset is an outcome-restricted sensitivity.

5.2 Spectral quantities

For each complete matrix X , columns with zero variance were removed and the remaining columns were standardized to zero mean and unit sample variance, producing Z . We formed the target correlation matrix

$$C = \frac{1}{N-1} Z^T Z \quad (4)$$

and sorted its non-negative eigenvalues $\lambda_1 \geq \dots \geq \lambda_P$. The primary summary was the PR effective dimension in Eq. 2. We also computed the entropy rank $\exp[-\sum_k p_k \log p_k]$, where $p_k = \lambda_k / \sum_l \lambda_l$, the fraction explained by principal component 1 and the number of components reaching 80% and 90% cumulative variance. Equation 3 follows from $\text{tr}(C^2) = P + P(P-1)\overline{r_{ij}^2}$. The PR effective dimension is especially sensitive to dominant eigenvalues, whereas entropy rank

weights the spectral tail more strongly. Both are continuous summaries of variance concentration, not algebraic or nonlinear manifold ranks.

The Marchenko–Pastur upper edge $(1 + \sqrt{P/N})^2$ was used as an asymptotic guide. The original panel analyses anchored mode counts on column-wise permutation parallel analysis, which preserves each score marginal while breaking cross-target alignment.[15] For direct column-standardized–residual comparison, we additionally drew the same 12,000 rows from each large matrix, generated 500 independent column-wise permutations, applied either no centering or two-way centering to both observed and permuted matrices, and counted observed eigenvalues above the rank-wise 95th percentile of the corresponding null. Because rank-wise comparisons do not control multiplicity, we also formed a 95% simultaneous envelope from the maximum studentized deviation over all eigenvalue ranks. Rank-wise counts are treated as descriptive; the simultaneous envelope controls the family-wise error rate. We repeated the count in five independent 100-permutation series to assess Monte Carlo stability. The 44-target analysis additionally used cluster-preserving and cluster-representative nulls; the latter retained three leading modes and controlled the inflated false-positive rate caused by chemical clustering. Effective-dimension intervals and the paired raw-to-residual difference for that panel resampled Butina clusters 500 times. DOCKSTRING used 100 molecule and 100 Bemis–Murcko scaffold-cluster replicates on each of five independently seeded 15,000-row supports. The primary matched experiment used a paired ligand bootstrap. These intervals are conditional on the observed target panels; chemical resampling does not represent target-selection uncertainty.

5.3 Main-effect removal

For complete matrices we computed the interaction residual in one closed-form step,

$$\Gamma = X - \bar{X}_{\cdot j} - \bar{X}_{i \cdot} + \bar{X}_{\cdot \cdot}, \quad (5)$$

where the bars denote column, row and grand means. We then column-standardized Γ and computed exactly the same spectral summaries. Because standardization already removes column offsets, the change from the column-standardized to residual spectra is driven principally by removal of the per-ligand common direction. The fraction of grand-centered unstandardized sums of squares retained by Γ was reported separately and was not used as a PR-dimension measure. Each row of Γ sums to zero. Subsequent non-zero column scaling preserves one exact linear

dependence, so $\text{rank}(\Gamma) \leq P - 1$ and the residual correlation spectrum contains one structural zero eigenvalue. For an additive model with independent, identically distributed residual noise, the non-zero residual spectrum is flat and the PR dimension approaches $P - 1$. We therefore treat $P - 1$ as a noise ceiling, not as a panel-size correction, and show target-subsampling curves for the residual estimand.

To test whether centering alone mechanically produced the observed residual spectra, we fit the additive model in Eq. 1 on a common 12,000-row support. For each of 500 realizations, residuals were drawn independently from zero-mean Gaussian distributions with the observed target-specific residual variances, then added to the fitted grand, ligand and target effects. We two-way centered each simulated matrix and recomputed PR dimension. The one-sided empirical probability was calculated with the finite-simulation correction $(1 + k)/(1 + 500)$, where k is the number of null PR dimensions no larger than the observed value.

5.4 Physicochemical-axis analyses

The full-panel and matched-support residualizations used seven RDKit descriptors: heavy-atom count, molecular weight, Labute accessible surface area, topological polar surface area, Crippen cLogP, rotatable-bond count and ring count. Descriptors were standardized. For the full anti-target panel, a separate ordinary least-squares model was fit for each target and residuals were assembled before spectral analysis. The nonlinear sensitivity used five-fold shuffled cross-fitting (seed 0) with one histogram gradient-boosting regressor per target (200 iterations, learning rate 0.05, maximum depth 4); no ligand’s fitted value came from a model trained on that ligand. The same protocol was repeated on the matched 74-by-6 support. The narrow-size sensitivity retained molecules with 17–22 heavy atoms, and ligand efficiency divided each score by heavy-atom count. A separate cached diagnostic of the shared score direction used molecular weight, cLogP, topological polar surface area, hydrogen-bond donor and acceptor counts, rotatable-bond count and molecular volume. Pocket volumes were read from the frozen 44-pocket structure summary; 36 targets had complete volumes. The reported univariate statistic correlates pocket volume with the absolute correlation between each target score and the per-ligand panel mean, not with a principal-component loading. The 13-descriptor pocket model was evaluated by leave-one-target-out prediction.

5.5 Preprocessing and experimental-noise sensitivities

For the antitarget panel we repeated the column-standardized analysis with target-median imputation ($r_{\text{PR}} = 1.835$), complete cases (9,297 ligands; 1.320), pairwise-complete Pearson correlations (1.788) and pairwise Spearman correlations (1.686). The complete-case estimate changes the chemical support and is not a like-for-like imputation test. Censoring sensitivities treated source-clipped zero cells as missing, removed every affected ligand or removed the most affected target decile. For interaction spectra we compared score-scale centering followed by standardization (9.303), standardization followed by centering (9.378), and robust median/interquartile-range scaling followed by centering (9.462). These analyses test analysis-pipeline sensitivity rather than define alternative primary endpoints.

For the primary 74-by-6 ChEMBL block, target-median imputation was supplemented by 100 posterior multiple imputations from chained Bayesian-ridge regressions. Imputed values were bounded by the observed range of each target. We recomputed both raw and residual PR dimensions and all downstream preference metrics in every completed matrix. Central 95% ranges across imputations describe missing-value sensitivity and are not interpreted as frequentist confidence intervals.

Different-document ChEMBL repeats for the same compound–target pair provided a conservative repeatability scale. From 279,012 paired records, the median absolute difference was 0.47 pChEMBL and the implied single-measurement Gaussian standard deviation was 0.493. For each of the six matched targets, this standard deviation was divided by that target’s experimental standard deviation. Independent Gaussian noise at the resulting target-specific scales was added to the standardized docking block in 2,000 simulations before recomputing PR dimension. This calibration includes inter-laboratory and condition variation; it is not a pure technical replicate variance or a complete experimental noise model.

5.6 Chemical- and target-composition sensitivities

Target-subsampling curves used 200 subsets without replacement at each reported size on one seeded 12,000-ligand support; the complete-panel points used all eligible ligands. Uniform sampling was paired with broad family-stratified sampling based on the versioned `data/target_families.csv` manifest. Stratified allocations were proportional to family size and included at least one target from every family whenever the requested subset size permitted. Both the

column-standardized and two-way-centered residual PR dimensions were recomputed after every draw.

Leave-one-target-out analysis used all 12,651 Docking-44 ligands and one seeded 15,000-row DOCKSTRING support; complete-panel estimates on the same support were plotted as references. The resulting minima, medians and maxima are composition sensitivities, not confidence intervals for a target superpopulation. Ligand-subsampling curves used 50 subsets. For the DOCKSTRING chemical-cluster sensitivity, Bemis–Murcko scaffolds were generated with RDKit; acyclic molecules were treated as singletons. One hundred cluster-bootstrap replicates were compared with 100 molecule-bootstrap replicates on each of five independent 15,000-molecule supports. The first support used the prespecified seed 0; the remaining support seeds were drawn reproducibly from that generator.

5.7 Expanded matched target-preference benchmark

The operational endpoint used the 691 observed exact-relation median ChEMBL cells in the 137-by-38 sparse overlap. For each ligand and every pair of observed targets, a prediction was correct when the more negative docking representation corresponded to the larger experimental pChEMBL value. Exact pChEMBL ties were excluded, leaving 2,522 comparisons. The primary statistic was the mean of per-ligand pairwise accuracies, giving each ligand equal weight; a predicted score tie received half credit and pair-weighted accuracy was secondary. The Vina energy sign convention—more negative predicts stronger binding—was fixed before examining the results.

Six operational representations were compared. Absolute Vina used the original scores. A docking target-prior baseline assigned every evaluation ligand the target means estimated from the 12,514 Docking-44 ligands outside the evaluation cohort. Column-standardized scores used the corresponding external target means and sample standard deviations. Two-way residual scores additionally removed each evaluation ligand’s row mean over all 44 docking targets and the external training grand mean, then divided by external residual target standard deviations. Missing evaluation scores were filled by target means from the same external fit. Thus, all docking transformation and imputation parameters were fitted without any of the 137 evaluation ligands. Relative to column standardization, this operational residualization is a constrained target rescaling plus a ligand row effect weighted by the inverse residual target scales, not an arbitrary learned reranker. An external experimental prior ranked target mean pChEMBL values

estimated from the 96,744 exact-relation ChEMBL ligands outside the evaluation cohort. A second, deliberately stringent diagnostic prior used target mean pChEMBL within the broadly profiled evaluation cohort, fitted with the evaluation ligand’s entire Bemis–Murcko scaffold cluster held out; it measures target and observation-process structure rather than a deployable docking rule.

Uncertainty was estimated by 5,000 ligand bootstraps and 5,000 Bemis–Murcko scaffold-cluster bootstraps over 95 matched clusters. Paired comparisons resampled per-ligand accuracy differences. For absolute, column-standardized and residual scores, 5,000 ligand-identity shuffles broke molecule-specific score–experiment correspondence while preserving target columns; 5,000 within-ligand outcome permutations supplied a second null. One-sided probabilities used the finite-permutation correction. Sensitivities used minimum absolute experimental differences of 0.10, 0.50 and 1.00 pChEMBL, minimum coverage of three, five or six observed targets, leave-one-target-out deletion, human binding assays, human binding K_i/K_d , pair weighting and removal of cells whose repeated measurements spanned more than one pChEMBL unit. Coverage-stratum intervals used the same scaffold bootstrap and are pointwise; the nested strata are not independent. The separate dense 74-by-6 analysis retained its 100 multiple-imputation sensitivity for spectral quantities and is reported in Supplementary Table S3.

5.8 Software and reproducibility

The new analyses and figures were run with Python 3.12.11, NumPy 1.26.4, pandas 2.3.2, SciPy 1.16.2, scikit-learn 1.7.2, Matplotlib 3.10.6 and RDKit 2025.03.6. The release’s `analysis/build_evidence.py` regenerates the centering ladder, preprocessing controls, experimental sensitivities, the expanded observed-pair benchmark, subsampling results and DTI associations in `results/evidence_summary.json`; it also embeds explicitly labelled summaries from the frozen upstream docking/rescoring stages. `analysis/make_figures.py` regenerates all main and supplementary figures. The core release reproduces the reported analysis from frozen matrices and summaries; it does not rerun receptor preparation, docking or RF-Score pose rescoring. A checksum manifest, pinned requirements, environment file, Dockerfile and Makefile provide input verification and one-command regeneration. Random seed 0 initializes all stochastic procedures; derived support and replicate seeds are stored in the evidence ledger.

OpenAI Codex and Anthropic Claude were used during manuscript preparation for code review, literature-search assistance and language editing. They were not treated as sources of

scientific evidence. The authors inspected the underlying data and code, verified the reported claims and citations, and retain responsibility for the work.

List of abbreviations

DTI, drug–target interaction; KNN, k-nearest neighbors; PC, principal component; RMT, random matrix theory; PR, participation ratio.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of data and materials

The source antitarget dataset is available with Nikitin et al.[10]; DOCKSTRING scores and poses are available from the DOCKSTRING release.[9] DAVIS and ChEMBL are publicly available from their respective source repositories. The derived matrices, frozen summaries, source manifest and analysis code are available without registration under an OSI-approved code license in the versioned GitHub repository and Zenodo archive.[25, 26] The exact submission package corresponds to GitHub tag v1.2.0. Upstream datasets retain their stated licenses and are identified by source URL, version and checksum in the manifest.

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

A.L. conceived the study, performed the analyses and drafted the manuscript. V.S. and M.F. contributed to study design and interpretation. All authors reviewed the manuscript and are responsible for the final submitted version.

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